

Question Number	Answer	Additional Guidance	Mark
1(a)	An answer that includes four of the following points. • one reference to safety (1) and three of the following points • use of cotton bud (1) • followed by use of stain/dye (1) • place cells (on slide) under coverslip (1) • use of high power of microscope (1)	e.g. bud into disinfectant/sterile/fresh bud/toothpick/wear gloves/ goggles/safe use of microscope/slides/careful use of bud/stain to prevent injury Accept toothpick/earbud/lollipop stick/glass rod/swab can piece together	(4)

Question Number	Answer	Additional Guidance	Mark
1(b)(i)	A drawing showing the following features: • drawing showing correct shape of cell and nucleus and nucleus in correct position (1) • any two correct labels (1)	Ignore other features drawn nucleus, cytoplasm, (plasma) membrane, nuclear membrane Ignore other labelled features unless specifically plant cell ones, then this mark negated	(2)

Question Number	Answer	Additional Guidance	Mark
1(b)(ii)	• reading of width using scale (1) • width calculated (1)	Example of calculation: 33/34/35 units So actual width is $33/34/35 \times 3 = 99/102/105 \text{ (}\mu\text{m)}$ so no marks for mp2 if other answers to mp1 unless 32 or 36, or 3.3, 3.4, 3.5 correct answer with no working gains both marks	(2)

Question Number	Answer	Additional Guidance	Mark
1(c)(i)	calculation of haemoglobin molecule volume	Example of calculation $4/3\pi r^3 = 1.3 \times 3.14 \times 2.5^3 = 65.45\text{nm}^3$ Other acceptable answers 65.5/63.81/63.8/65.42/65.4/63.78	(1)

Question Number	Answer	Additional Guidance	Mark
1(c)(ii)	A calculation showing the following steps: - volume of red blood cell ÷ volume of haemoglobin molecule (1) - conversion and standard form (1) OR - conversion of {red blood cell volume, from μm^3 to nm^3 / Hb molecule radius, 2.5 nm to 0.0025 μm / Hb volume calculated from nm^3 to μm^3} (1) - division of rbc volume in $\{\text{nm}^3 \div \mu\text{m}^3\}$ by calculated haemoglobin molecule volume in $\{\text{nm}^3 / \mu\text{m}^3\}$ (1)	Example of calculation 80 ÷ (answer from 1ci) e.g.= 80 ÷ 65.45 = {1.222/1.22/1.2} 1.2/1.22 × 10⁹ 80 μm^3 to 8 × 10¹⁰ nm^3 / Hb molecule radius, 2.5 nm to 0.0025 μm e.g {80 000 000 000 / 8 × 10¹⁰} ÷ 65.44 = 1 222 493 887 = 1.2/1.22 × 10⁹	(2)

Question Number	Answer	Additional Guidance	Mark
1(c)(iii)	A comparative description that includes any two of the following: • one difference described for plant cell and RBC (1) • another difference described for plant cell and RBC (1)	e.g. plant cell has nucleus, RBC does not allow one mark if both statements correct but not comparative plant cell: nucleus/nucleolus/vacuole/large(r) vacuole/chloroplasts/cell wall/regular shape/bigger/colour accept converse	(2)